KID65001AP/AF~ KID65004AP/AF

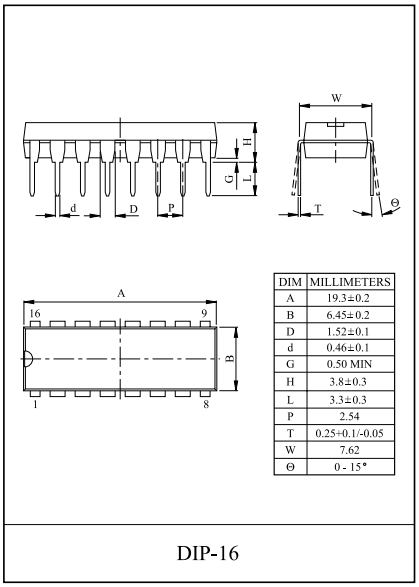
BIPOLAR LINEAR INTEGRATED CIRCUIT

7 CIRCUIT DARLINGTON TRANSISTOR ARRAY

FEATURES

- Output Current : 500mA Max.
- High Sustaining Voltage Outputs : 50V Min.
- Output Clamp Diodes.
- Inputs Compatible With Various Types of Logic.
- PKG Type AP : DIP-16Pin, DIP-16Pin(1) AF : FLP-16Pin

TYPE	INPUT RESISTOR	DESIGNATION
KID65001AP/AF	No (External)	General Purpose
KID65002AP/AF	Zener Diode 7V+10.5kΩ	14 25V P-MOS
KID65003AP/AF	2.7kΩ	TTL, 5V C-MOS
KID65004AP/AF	10.5kΩ	6 15V P-MOS, C-MOS



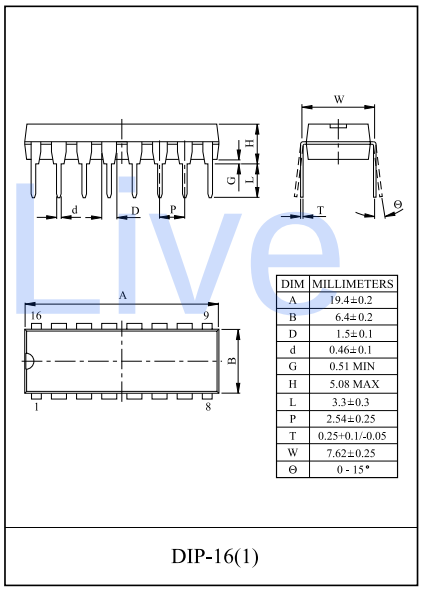
DIM	MILLIMETERS
A	19.3±0.2
B	6.45±0.2
D	1.52±0.1
d	0.46±0.1
G	0.50 MIN
H	3.8±0.3
L	3.3±0.3
P	2.54
T	0.25+0.1/-0.05
W	7.62
Θ	0 - 15°

DIP-16

DESCRIPTION:

The KID65001AP/AF Series are high-voltage, high-current darlington transistor array comprised of seven NPN darlington pairs. All units feature internal clamp diodes for switching inductive loads.

Datasheet.Live



DIM	MILLIMETERS
A	19.4±0.2
B	6.4±0.2
D	1.5±0.1
d	0.46±0.1
G	0.51 MIN
H	5.08 MAX
L	3.3±0.3
P	2.54±0.25
T	0.25+0.1/-0.05
W	7.62±0.25
Θ	0 - 15°

DIP-16(1)

MAXIMUM RATINGS (Ta=25℃, unless otherwise noted)

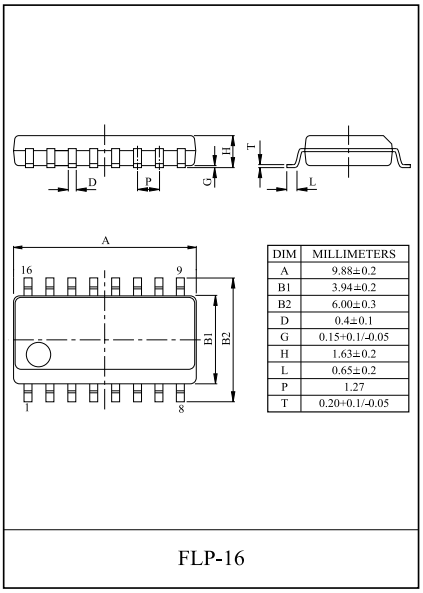
CHARACTERISTIC		SYMBOL	RATING	UNIT
Output Sustaining Voltage		$V_{CE(SUS)}$	50	V
Output Current		I_{OUT}	500	mA/ch
Input Voltage		I_{IN} 1)	-0.5 +30	V
Input Current		I_{IN} 2)	25	mA
Clamp Diode	Reverse Voltage	V_R	50	V
	Forward Current	I_F	500	mA
GND Terminal Current		I_{GND}	2.8	A
Power Dissipation	AP	P_D	1.47	W
	AF		0.54 /0.63 3)/1.25 4)	W
Operating Temperature		T_{opr}	-40 85	℃
Storage Temperature		T_{stg}	-55 150	℃

1) Except KID65001AP/AF

2) Only KID65001AP/AF

3) On PCB(30×30×1.6mm, Cu 50%)

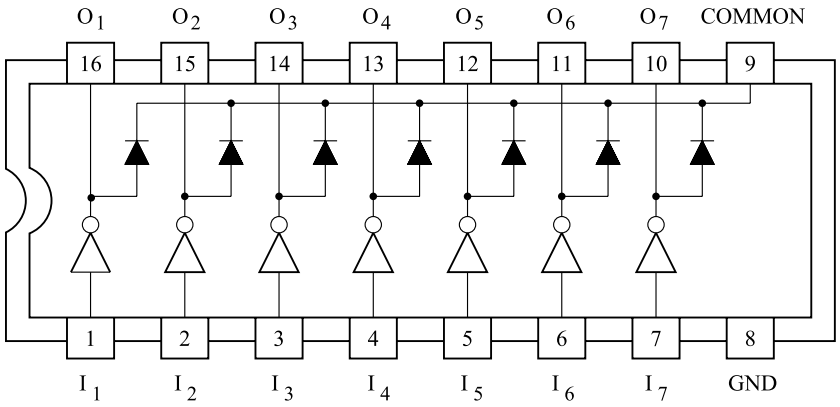
4) On PCB (Test Board : JEDEC 2s2p)



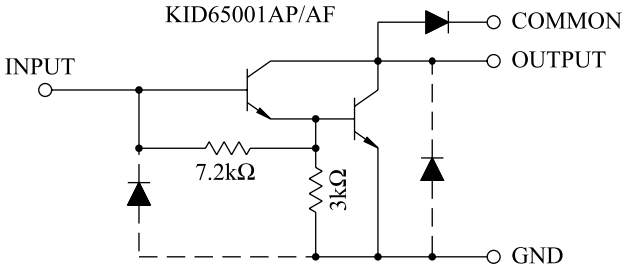
DIM	MILLIMETERS
A	9.88±0.2
B1	3.94±0.2
B2	6.00±0.3
D	0.4±0.1
G	0.15+0.1/-0.05
H	1.63±0.2
L	0.65±0.2
P	1.27
T	0.20+0.1/-0.05

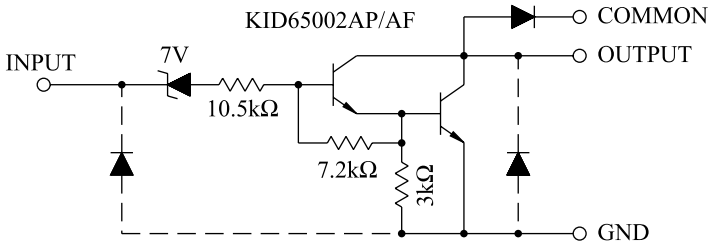
FLP-16

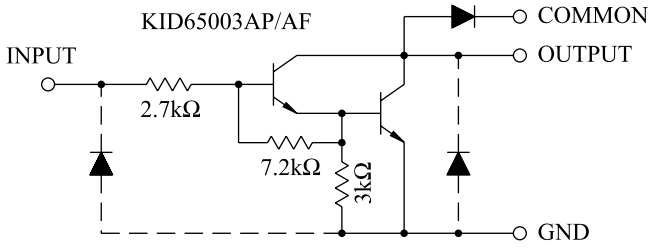
PIN CONNECTION (TOP VIEW)

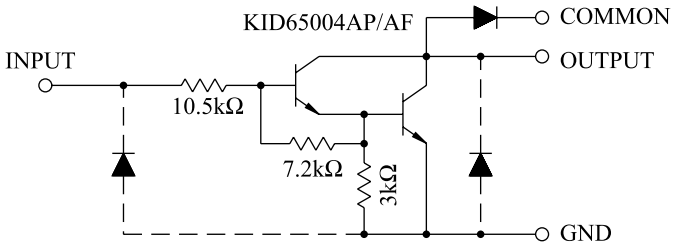


SCHEMATICS (EACH DRIVER)









RECOMMENDED OPERATING CONDITIONS (Ta=-40 85℃)

CHARACTERISTIC		SYMBOL	CONDITION	MIN.	TYP.	MAX.	UNIT
Output Sustaining Voltage		$V_{CE(SUS)}$		0	-	50	V
Output Current		I_{OUT} (AP, T_a=85℃)	T_{PW}=25ms, DF=10%, 7 Circuits	0	-	370	mA
			T_{PW}=25ms, DF=30%, 7 Circuits	0	-	200	
		I_{OUT} (AF, T_a=85℃)	T_{PW}=25ms, DF=10%, 7 Circuits	-	-	290	mA
			T_{PW}=25ms, DF=30%, 7 Circuits	-	-	150	
Input Voltage		V_{IN}	Except KID65001AP/AF	0	-	30	V
Input Current		I_{IN}	Only KID65001AP/AF	0	-	5	mA
Clamp Diode Reverse Voltage		V_R		-	-	50	V
Clamp Diode Forward Current		I_F		-	-	400	mA
Power Dissipation	AP	P_D	Ta=85℃	-	-	0.76	W
	AF		Ta=85℃	0.28 / 0.32* / 0.65**			

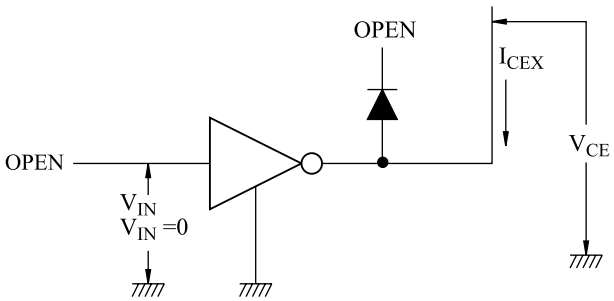
* On PCB (30×30×1.6mm, Cu 50%)
** On PCB (Test Board : JEDEC 2s2p)

ELECTRICAL CHARACTERISTICS (Ta=25℃, unless otherwise noted)

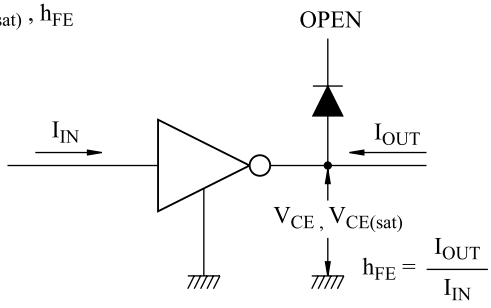
CHARACTERISTICS		SYMBOL	TEST CIRCUIT	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Output Leak Current		I_{CEX}	1	V_{CE}=50V, Ta=25℃	-	-	50	μA
				V_{CE}=50V, Ta=85℃	-	-	100	
	KID65002AP/AF			V_{CE}=50V, V_{IN}=6V	-	-	500	
	KID65004AP/AF			V_{CE}=50V, V_{IN}=1V	-	-	500	
Collector-Emitter Saturation Voltage		$V_{CE(sat)}$	2	I_{OUT}=350mA, I_{IN}=500 μA	-	1.3	1.6	V
				I_{OUT}=200mA, I_{IN}=350 μA	-	1.1	1.3	
				I_{OUT}=100mA, I_{IN}=250 μA	-	0.9	1.1	
Input Current	KID65002AP/AF	$I_{IN(ON)}$	3	V_{IN}=17V	-	0.82	1.25	mA
	KID65003AP/AF			V_{IN}=3.85V	-	0.93	1.35	
	KID65004AP/AF			V_{IN}=5V	-	0.35	0.5	
				V_{IN}=12V	-	1.0	1.45	
		$I_{IN(OFF)}$	4	I_{OUT}=500 μA, Ta=85℃	50	65	-	μA
Input Voltage	KID65002AP/AF	$V_{IN(ON)}$	5	V_{CE}=2V, I_{OUT}=300mA	-	-	13	V
	KID65003AP/AF			V_{CE}=2V, I_{OUT}=200mA	-	-	2.4	
				V_{CE}=2V, I_{OUT}=250mA	-	-	2.7	
				V_{CE}=2V, I_{OUT}=300mA	-	-	3.0	
	KID65004AP/AF			V_{CE}=2V, I_{OUT}=125mA	-	-	5.0	
				V_{CE}=2V, I_{OUT}=200mA	-	-	6.0	
				V_{CE}=2V, I_{OUT}=275mA	-	-	7.0	
				V_{CE}=2V, I_{OUT}=350mA	-	-	8.0	
	KID65002AP/AF	$V_{IN(OFF)}$		-	0	-	7.4	
	KID65003AP/AF			-	0	-	0.7	
	KID65004AP/AF			-	0	-	1.0	
DC Current Transfer Ratio		h_{FE}	2	V_{CE}=2V, I_{OUT}=350mA	1000	-	-	
Clamp Diode Reverse Current		I_R	6	V_R=50V, Ta=25℃	-	-	50	μA
				V_R=50V, Ta=85℃	-	-	100	
Clamp Diode Forward Voltage		V_F	7	I_F=350mA	-	-	2.0	V
Input Capacitance		C_{IN}			-	15	-	pF
Turn-ON Delay		t_{ON}	8	V_{OUT}=50V, R_L=163Ω	-	0.1	-	μs
Turn-OFF Delay		t_{OFF}			-	0.2	-	

TEST CIRCUIT

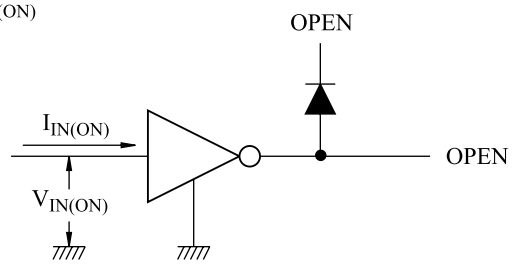
1. I_{CEX}



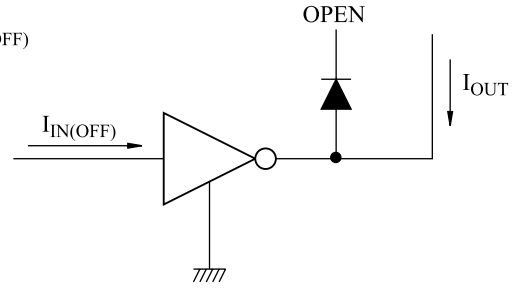
2. $V_{CE(sat)}$, h_{FE}



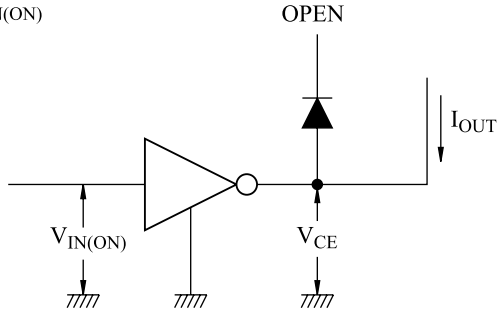
3. $I_{IN(ON)}$



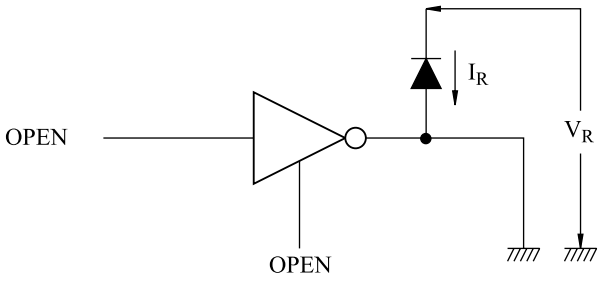
4. $I_{IN(OFF)}$



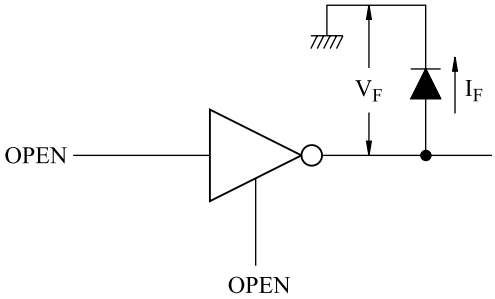
5. $V_{IN(ON)}$



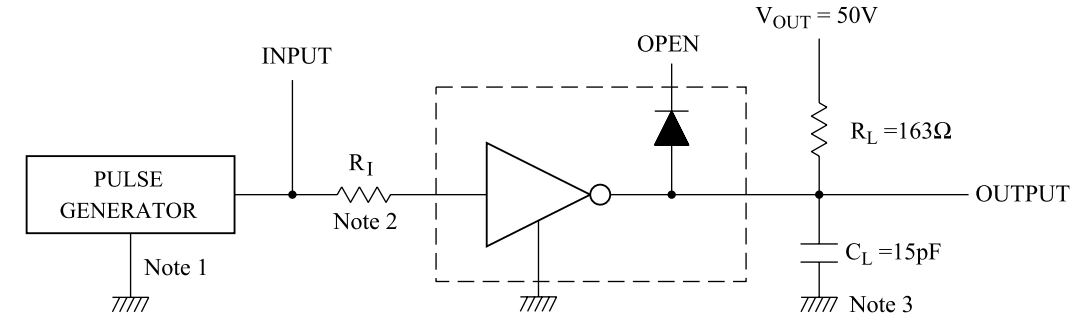
6. I_R

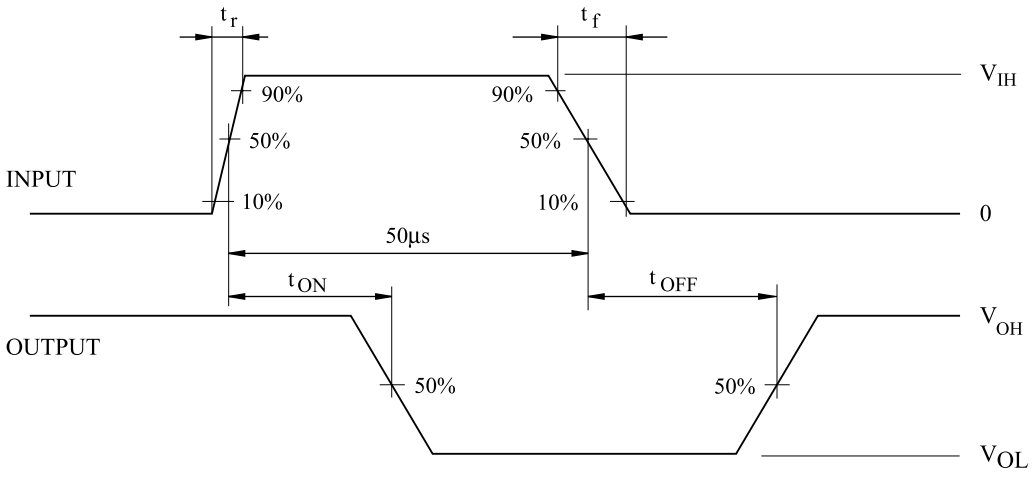


7. V_F



8. t_{ON} , t_{OFF}





Notes : 1. Pulse Width 50 μs, Duty Cycle 10%
Output Impedance 50Ω, t_r≦5ns, t_f≦10ns

2. See below

Input Conditions

Type Number	R_I	V_{IH}
KID65001AP/AF	2.7kΩ	3V
KID65002AP/AF	0	13V
KID65003AP/AF	0	3V
KID65004AP/AF	0	8V

3. C_L includes probe and Jig capacitance.

Fig. 1 I_{OUT} - DUTY CYCLE

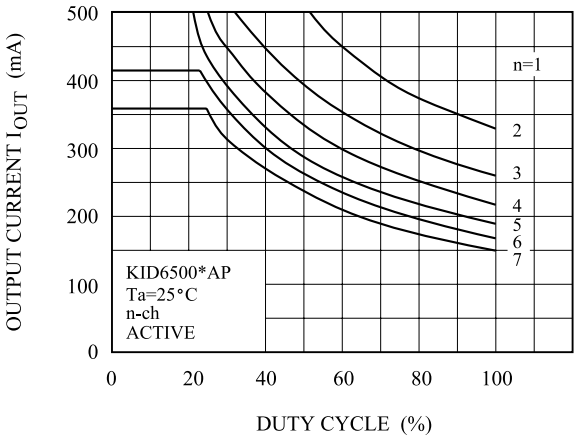


Fig. 2 I_{OUT} - DUTY CYCLE

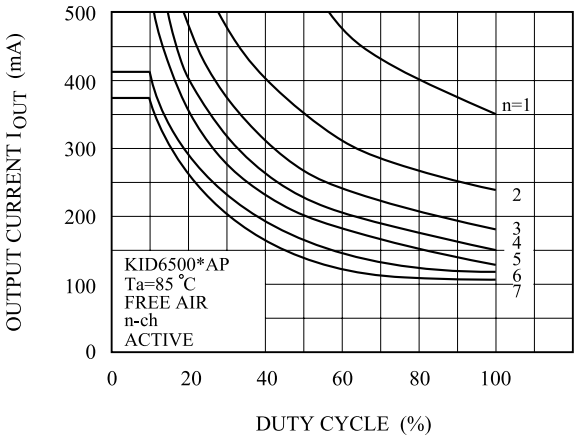


Fig. 3 I_{OUT} - DUTY CYCLE

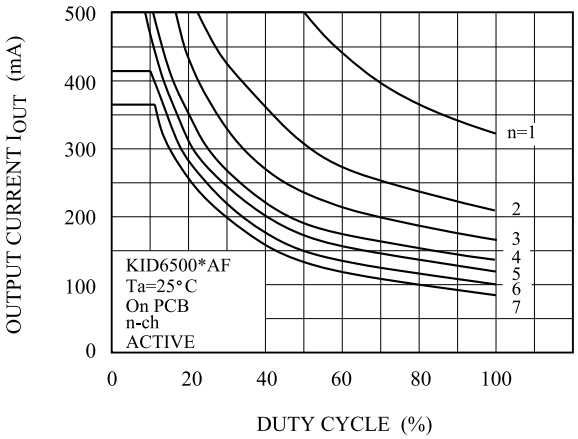


Fig. 4 I_{OUT} - DUTY CYCLE

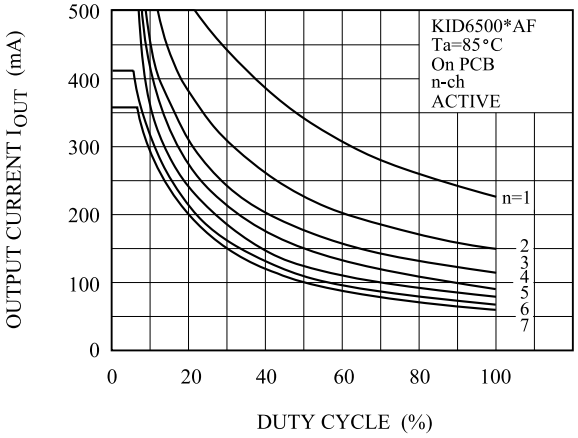


Fig. 5 I_{IN} - V_{IN}

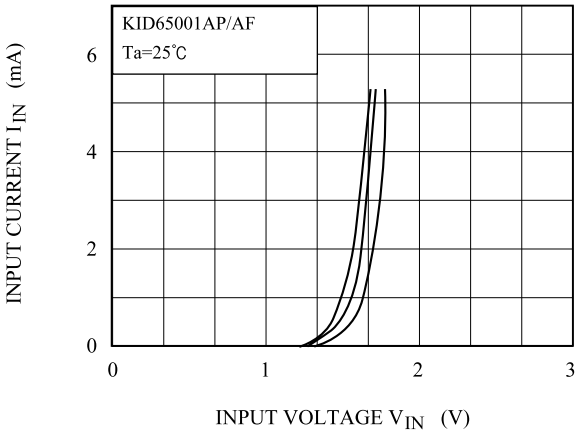


Fig. 6 I_{IN} - V_{IN}

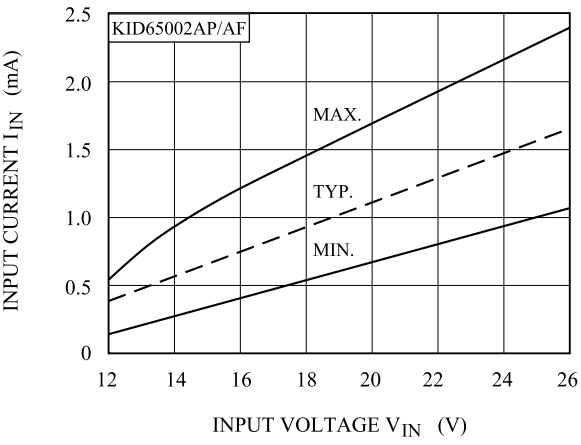


Fig. 7 I_{IN} - V_{IN}

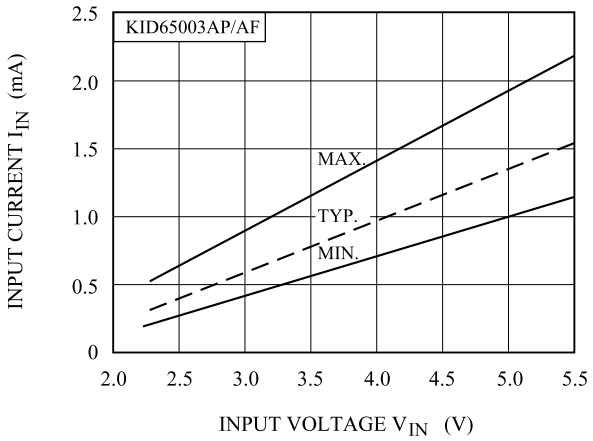


Fig. 8 I_{IN} - V_{IN}

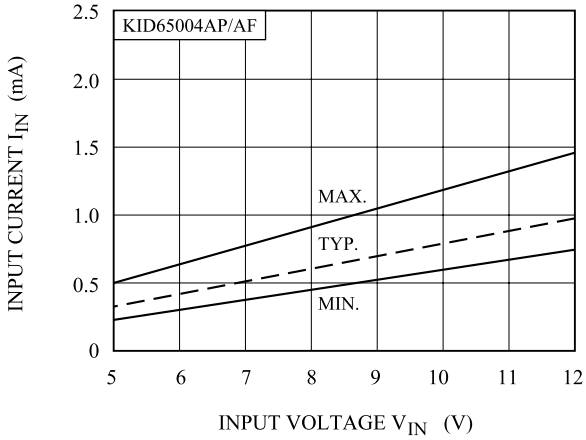


Fig. 9 I_{OUT} - $V_{CE(sat)}$

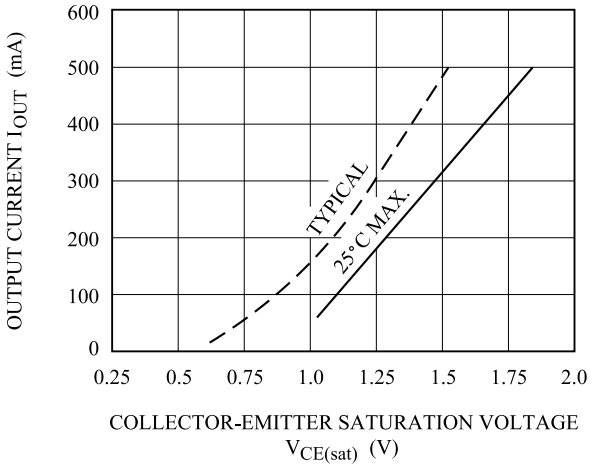


Fig. 10 h_{FE} - I_{OUT}

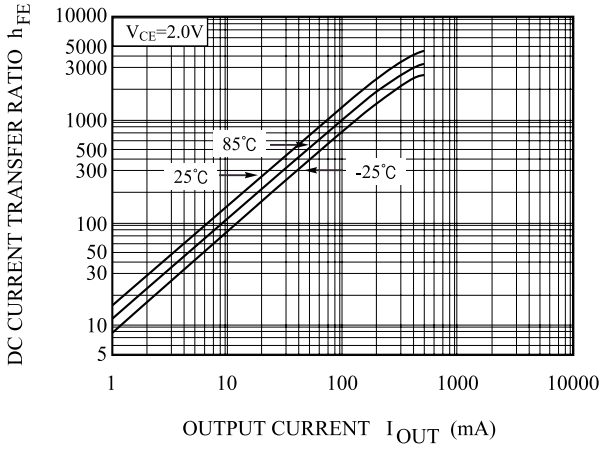


Fig. 11 I_F - V_F

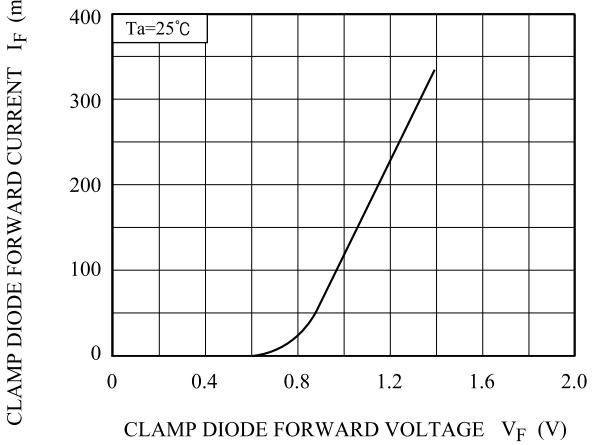


Fig. 12 P_D - Ta

